The Seam Between Day and Night

Teacher copy (everything in the student copy, plus answers and teaching notes)

The following examines WASP-121 b, a planet so close to its star that one hemisphere faces permanent day and the other permanent night, and how astronomers have begun to read the differences along the boundary between them.

The article, marked for rhetorical devices (one color throughout):

1 Eight hundred and fifty-eight light-years from Earth, a planet called WASP-121 b circles its star about once every thirty hours. It orbits so close that the star fills its sky. The heat has locked it in place, one face turned forever toward the fire, the other turned forever toward the dark **[Antithesis]**, so the planet has no sunrise and no sunset, only a permanent day on one side and a permanent night on the other.

2 Between those two hemispheres runs a seam **[Metaphor]**, and a team of astronomers at the Max Planck Institute for Astronomy has now read it edge by edge. On the day side the air reaches about 2,500 degrees Celsius, hot enough to turn metal and rock to vapor; on the night side it falls to around 700, still glowing but darker, a difference of more than 1,700 degrees across a single world. The boundary, the line astronomers call the terminator, is not one line here but two: a morning edge, where the dark turns toward light, and an evening edge, where the light fails into dark.

3 Reading a seam on a world that far away takes patience and a trick of geometry. As the planet crosses the face of its star, from Earth a small, regular eclipse, it turns by about thirty degrees, so the starlight that filters through its air passes through one longitude after another in sequence. That light carries a record of what it passed through. Split into its colors, it shows which gases lie along each edge and how hot they are. The two edges do not match.

4 The evening edge swallows more starlight than the morning edge. The reason is wind. **[Telegraphic sentence]** Superheated gas pours off the day side on a steady eastward current and piles into the evening air; the air swells with the heat, widens, and blocks more light. The evening edge is also drier. There the temperature runs high enough to tear water molecules apart into the atoms that make them, hydrogen and oxygen, so less water survives to be seen. In its place the trace of carbon monoxide grows stronger toward the evening, a tougher molecule that holds together where water cannot.

5 What the measurements do not fully explain is how large the difference is. The models of these atmospheres predict a gentler imbalance than the telescope found, which means something in the evening air, some haze or cloud the equations do not yet carry, is still unaccounted for. The planet was found in 2015, the first beyond our own system shown to hold water and to wrap itself in a warm upper layer. A decade later it is read longitude by longitude: its locked face, its eastward wind, its swelling evening edge, its vanishing water and its surviving carbon monoxide **[Parallelism]**, a single circling object that turns out to be made of differences, and not yet wholly known.

Devices, and why they are here:

Antithesis (paragraph 1): “one face turned forever toward the fire, the other turned forever toward the dark”
Sets the planet's permanent dayside against its permanent nightside in a single balanced clause, fixing the one opposition the whole piece explores: a world split into two unmoving halves.

Metaphor (paragraph 2): “a seam”
Calling the day-night boundary a seam turns an abstract line of physics into something a reader can picture being opened and read, and quietly frames the planet as a thing stitched from two unlike sides.

Telegraphic sentence (paragraph 4): “The reason is wind.”
After a long sentence about which edge absorbs more light, three blunt words deliver the cause and let the explanation that follows expand against them. The brevity is the emphasis.

Parallelism (paragraph 5): “its locked face, its eastward wind, its swelling evening edge, its vanishing water and its surviving carbon monoxide”
The repeated 'its' phrases pile the planet's features into one accumulating list whose growing length enacts the complexity it names, closing on the object itself rather than on any moral drawn from it.

AP-style multiple choice:

1. The relationship between the third paragraph and the fourth paragraph is best described as one in which the writer first

A) raises a doubt about the measurements, then resolves it with new evidence
B) explains the method used to read the planet's atmosphere, then gives the physical cause of the difference that method revealed
C) describes the planet's discovery, then predicts what future telescopes will find
D) compares the planet to Jupiter, then explains why the comparison breaks down

2. In the final paragraph, the phrase 'not yet wholly known' primarily serves to

A) warn that the telescope's measurements are probably mistaken
B) imply that astronomers have lost interest in the planet
C) concede that the planet's atmosphere resists complete explanation even after close study
D) claim that the planet can never be studied any further

Multiple choice answer key:

1. Answer B. The third paragraph lays out how starlight read during the planet's transit reveals each edge of its atmosphere, and the fourth paragraph supplies the cause, an eastward wind that swells and dries the evening edge; the keyed pairing tracks that move from method to mechanism. The discovery-and-prediction option fails because the discovery is mentioned only in the fifth paragraph and no forecast about future telescopes appears, a true detail tied to the wrong span. The doubt-and-resolution option imports a skepticism the passage raises only at the end, and about the models rather than the measurements. The Jupiter option names a comparison the passage never makes.

2. Answer C. The phrase caps a paragraph that admits the models predict a smaller imbalance than was observed, so the keyed option matches the writer's candid, measured tone about the limits of present knowledge. The mistaken-measurements option inverts the point: the passage faults the models, not the data. The lost-interest option imports an attitude the text never states, since the planet is described as freshly and closely read. The never-again option overreaches into an absolute the passage contradicts by showing the work is ongoing.

Discussion questions:

1. The writer accumulates physical detail about the planet but never says why the discovery matters or draws a lesson from it. What does the piece gain, and what might it give up, by withholding any judgment of significance?

2. Several times the writer states plainly what is not known, ending on a planet that is 'not yet wholly known.' How does admitting the limits of the account affect your trust in the writer?

Sample strong discussion responses:

1. By refusing to tell the reader the discovery is important, the writer lets the facts carry their own weight: the temperature gap, the wind, the dissolving water are allowed to be striking on their own rather than because a sentence insists they are. The gain is trust, since nothing is being oversold and the reader feels shown rather than told. What it gives up is direction. A reader who wants to know why an alien atmosphere should matter to them is left to supply that, because the piece stops at accurate description and declines the moral.

2. The repeated admissions, that the models predict less than was seen, that some cloud is unaccounted for, that the planet is not yet wholly known, make the writer more believable, not less. A voice that claimed to understand everything would invite suspicion; one that marks the edge of its own knowledge sounds like it is reporting honestly. The candor also mirrors how science actually works, as a record still open, so the unanswered questions read as part of the subject rather than as a failure of the account.

Writing prompt (Q2 rhetorical analysis):

Read the passage carefully. Then write an essay that analyzes the rhetorical choices the writer makes to convey both the strangeness of WASP-121 b and the limits of what is currently known about it. Consider how the writer's accumulation of physical detail, contrast between short and long sentences, and admissions of uncertainty work together.

Common misconceptions to watch for:

- That a tidally locked planet does not rotate. It rotates exactly once each orbit, which is why one face stays toward the star, and that slow turn is precisely what lets astronomers read different longitudes of its air during a single transit.
- That 'a seam' means the planet has a literal stitched line. It is a metaphor for the terminator, the broad boundary region between the permanent dayside and nightside, not a visible crack.
- That the writer's closing admission of uncertainty means the measurements are unreliable. The passage says the models predict a smaller difference than was observed, which points to incomplete theory, not faulty data.

AP exam alignment:

This opener is built in the AP English Language source-passage mold. The Long Look register models Rhetorical Situation (Skill Category 1): an absorbed, impersonal observer; a curious general reader; and an exigence of illumination rather than dispute. The accumulation structure, building edge by edge to a final piled list, gives Reasoning and Organization (Skill Category 5) a clear line to trace, targeted by the first multiple-choice item. The short-against-long rhythm, the single controlling metaphor, and the candid admissions of uncertainty give Style (Skill Category 7) concrete material, targeted by the second item. The two questions test different reading skills. The free-response prompt is a Q2 rhetorical analysis, the fitting choice because the passage renders a phenomenon rather than arguing a contestable position; no Q3 argument is stapled on, since the piece advances no debatable claim to defend, challenge, or qualify.